Arista Access Node WFD-ACC-71 Hardware/Environmental Fault Resolution SOP

Node: WFD-ACC-71 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1808

Description

This Standard Operating Procedure (SOP) outlines the steps required to resolve hardware/environmental faults on Arista access node WFD-ACC-71 at the Whitefield POP location. The failure domain in question is related to ASIC memory parity errors, line card crashes, and power supply redundancy loss, resulting in an ALARM_1808 minor severity alarm. The Arista ES-4000 series access nodes, specifically the 4000-32C model, utilize a combination of ASIC-based forwarding and line card-based interface aggregation. The memory parity errors and line card crashes can be attributed to the inherent complexity of these architectures, where multiple components interact to facilitate high-speed forwarding. Furthermore, the loss of power supply redundancy can be caused by various factors, including power grid instability, inadequate cabling, or faulty power supply units. Understanding these underlying mechanisms is crucial for effective troubleshooting and resolution. The following SOP outlines the necessary steps to diagnose and resolve the hardware/environmental fault, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Verify that the Whitefield POP location has been dispatched with a team of skilled engineers and technicians.

- Check the redundant link status to ensure that it is operational and not contributing to the fault.

- Confirm that the power supply units are functioning correctly and that there are no signs of overheating or power issues.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Error	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal forwarding-engine, Expected Output: Memory parity error detected on ASIC , Description: This command isolates the ASIC memory parity error by displaying the system's internal forwarding engine status. The expected output indicates that a memory parity error has been detected on the specified ASIC ID.
```

```
Command: show module , Expected Output: Line card is not responding, Description: This command identifies the line card crash by displaying the status of the specified module ID. The expected output indicates that the line card is not responding, confirming the crash.

Command: show power-supply, Expected Output: Power supply unit is not functioning, Description: This command diagnoses the power supply redundancy loss by displaying the status of the power supply units. The expected output indicates that the specified power supply unit is not functioning, contributing to the redundancy loss.
```

Resolution Steps

Step 1: Step: Replace Faulty ASIC, Command: no forwarding-engine replace, Description: This step involves replacing the faulty ASIC with a new one. The command no forwarding-engine replace is used to remove the faulty ASIC from the forwarding engine, and then replace it with a new one. The command is executed in the system's internal configuration mode.

Step 2: Step: Reboot Line Card, Command: reboot module , Description: This step involves rebooting the line card to resolve the crash. The command reboot module is used to reboot the specified line card. This command is executed in the system's operational mode.

Step 3: Step: Replace Faulty Power Supply Unit, Command: replace power-supply , Description: This step involves replacing the faulty power supply unit to resolve the redundancy loss. The command replace power-supply is used to replace the specified power supply unit with a new one. This command is executed in the system's operational mode.

Post-Check Verification

```
Command: show system internal forwarding-engine, Expected Output: No memory parity errors detected, Description: This command verifies the resolution of the ASIC memory parity error by displaying the system's internal forwarding engine status. The expected output indicates that no memory parity errors are detected.

Command: show module , Expected Output: Line card is responding, Description: This command verifies the resolution of the line card crash by displaying the status of the specified module ID. The expected output indicates that the line card is responding, confirming the fix.

Command: show power-supply, Expected Output: Power supply unit is functioning, Description: This command verifies the resolution of the power supply redundancy loss by displaying the status of the power supply units. The expected output indicates that the specified power supply unit is functioning, confirming the fix.
```

Roll-back Procedures

- Step: Revert Faulty ASIC, Command: forwarding-engine revert, Description: This step involves reverting the faulty ASIC to its original state. The command forwarding-engine revert is used to restore the faulty ASIC to its original state. This command is executed in the system's internal configuration mode.

- Step: Revert Line Card Configuration, Command: undo reboot module , Description: This step involves reverting the line card configuration to its original state. The command undo reboot module is used to restore the line card configuration to its original state. This command is executed in the system's operational mode.

- Step: Revert Power Supply Unit Replacement, Command: undo replace power-supply , Description: This step involves reverting the power supply unit replacement to its original state. The command undo

replace power-supply is used to restore the power supply unit to its original state. This command is executed in the system's operational mode.